

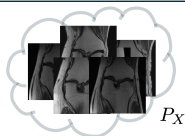
ERASMUS+ INTERNATIONAL PHD SUMMER SCHOOL 2025
Mathematics and Machine Learning for image analysis
Lecture 2 - Optimization vs. Sampling

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Bayesian inverse problems



- Bayesian inference

$$p_{X|Y}(x|y) = \frac{p_{Y|X}(y|x)p_X(x)}{p_Y(y)}$$

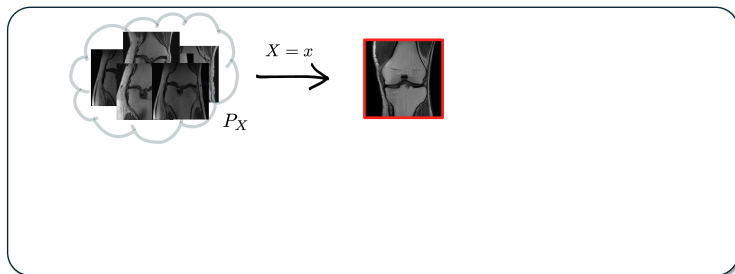
- Can be seen as **logic with uncertainty**.

- The likelihood $p_{Y|X}$ is often known due to the image formation process

$$p_{Y|X}(y|x) \propto \exp\left(-\frac{\|\mathcal{A}(x) - y\|^2}{2\sigma^2}\right)$$

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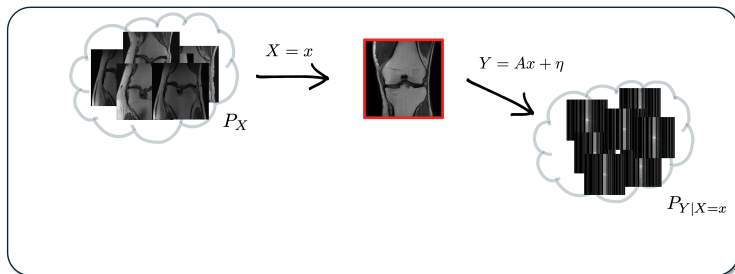
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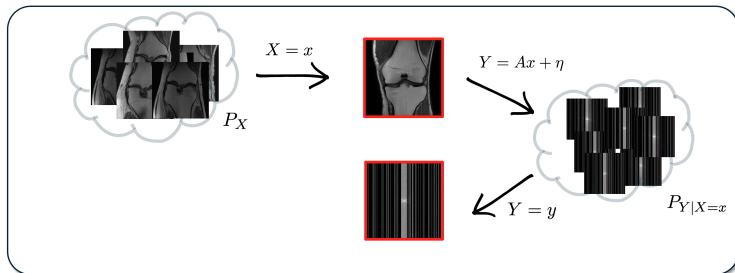
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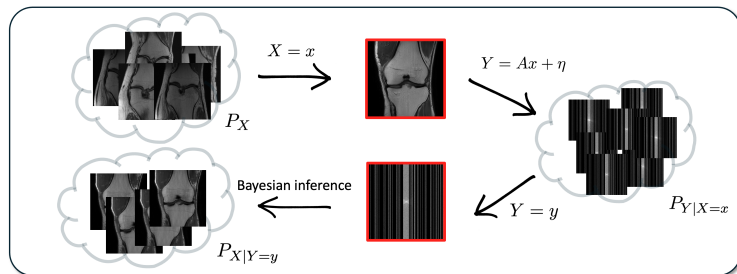
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Bayesian point estimators

- There are a number of Bayesian estimators that are used to compute point estimates from the posterior

$$\hat{x}(y) \in \operatorname{argmin}_z \int_{\mathcal{X}} \ell(x, z) p_{X|Y=y}(x) \, dx,$$

depending of the choice of the loss function $\ell(x, z)$.

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$$\hat{x}(y) = \operatorname{argmax}_x p_{X|Y=y}(x) = \operatorname{argmin}_x -\log p_{X|Y=y}(x)$$

- The squared loss leads to the posterior expectation, or minimum mean squared estimate (MMSE)

$$\bar{x}(y) = \int_{\mathcal{X}} x p_{X|Y=y}(x) \, dx = \mathbb{E}_{X|Y=y}[x]$$

Example: Total variation regularized Gaussian image denoising

- Let us assume we have a prior and likelihood that take the form of a Gibbs distribution [Boltzmann 1868], [Gibbs 1889]

$$p_X(x) \propto \exp(-R(x)), \quad p_{Y|X}(y|x) \propto \exp(-D(x, y)),$$

with R the regularizer and D the data fidelity term, which are here given by

$$R(x) = \lambda \sum_{i,j} |x_{i+1,j} - x_{i,j}| + |x_{i,j+1} - x_{i,j}|, \quad D(x, y) = \frac{\|x - y\|^2}{2\sigma^2}$$

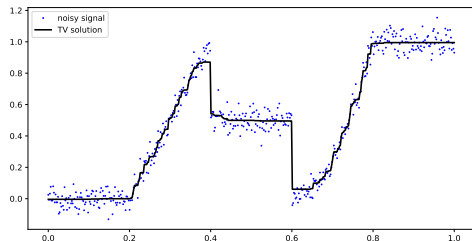
- In summary, the negative log posterior is given by

$$-\log p_{X|Y}(x|y) \propto E(x) := R(x) + D(x, y)$$

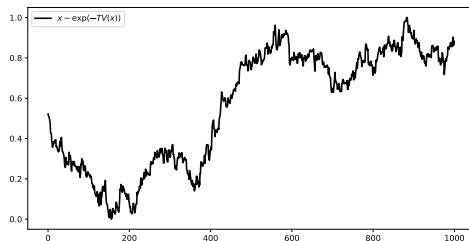
Regularizer vs. prior

The behavior of the total variation as a **regularizer** and a **prior** can be very different.

- It is well-known that using $R(x)$ as a regularization term (here in 1D) leads to piecewise constant signals [Chambolle, Lions, '97].



- This is in contrast to actual samples $x \sim p_x(x)$ which can be obtained for 1D TV in linear time via a Levy process [Bohra, et al., '23]



MAP estimation

- ▶ MAP estimation (minimization)

$$\hat{x} = \operatorname{argmin}_{x \in \mathcal{X}} R(x) + \frac{1}{2\sigma^2} \|x - y\|_2^2.$$

- ▶ The optimal solution is given by the proximal map [Moreau '62]

$$\hat{x} = (I + \sigma^2 \partial R)^{-1}(y) := \operatorname{prox}_{\sigma^2 R}(y),$$

which can be written as **one** gradient step on the

infimal-convolution of R with a quadratic function $\frac{\|\cdot\|_2^2}{2\sigma^2}$ (called Moreau envelope)

$$\hat{x} = y - \sigma^2 \nabla \hat{R}_{\sigma^2}(y), \quad \hat{R}_{\sigma^2} = (R \square \frac{\|\cdot\|_2^2}{2\sigma^2}),$$

also with a nice representation of its gradient

$$\nabla \hat{R}_{\sigma^2}(y) = \frac{y - \operatorname{prox}_{\sigma^2 R}(y)}{\sigma^2}.$$

- ▶ MMSE estimation (expectation)

$$\bar{x} = \frac{1}{Z} \int_{\mathcal{X}} x \exp \left(-R(x) - \frac{1}{2\sigma^2} \|x - y\|^2 \right) dx, \quad \bar{x} = \mathbb{E}[x|y],$$

which can be written in the form of **one** gradient step on the **soft-infimal-convolution** of $\exp(-R)$ with a Gaussian $\exp(-\frac{\|\cdot\|_2^2}{2\sigma^2})$

$$\bar{x} = y - \sigma^2 \nabla \bar{R}_{\sigma^2}(y), \quad \bar{R}_{\sigma^2} = -\log \left(\exp(-R) * \exp \left(-\frac{\|\cdot\|_2^2}{2\sigma^2} \right) \right).$$

- ▶ This formula is known as Tweedie's formula [Robbins '56], [Miyasawa '60].
- ▶ The negative gradients $-\nabla \bar{R}_{\sigma^2}$ are also known as score functions in denoising diffusion models at diffusion time $t = \sigma^2/2$.

Diffusion at absolute zero

- ▶ Let's introduce a temperature parameter $T > 0$

$$\bar{R}_{\sigma^2}^T(y) = -T \log \left(\int \exp \left(\frac{-R(x) - \frac{\|x-y\|_2^2}{2\sigma^2}}{T} \right) dx \right)$$

- ▶ One can show that as $T \rightarrow 0^+$, the **soft-infimal convolution** $\bar{R}_{\sigma^2}^T(y)$ converges to the **infimal convolution** $\hat{R}_{\sigma^2}(y)$.
- ▶ As a consequence, the “Tweedie-prox” becomes a “Moreau-prox”.
- ▶ This gives rise to a “Diffusion at absolute zero” [Habring, Falk, Zach, P. '25], as an alternative to the standard denoising diffusion models, where the score can be computed based on the proximal map.

Algorithms

- ▶ In practice, we usually do not have direct access to the gradients of the (soft-) infimal-convolutions to compute the MAP or MMSE in one step.
- ▶ The previous formulas can only be applied to image denoising.
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- ▶ The most basic optimization algorithm to compute the MAP estimate is gradient descent (GD).

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$$x^{k+1} = x^k - \tau \nabla E(x^k)$$

where $\tau > 0$ is the step size.

- ▶ The sampling analogue is the unadjusted Langevin algorithm (ULA)

$$X^{k+1} = X^k - \tau \nabla E(X^k) + \sqrt{2\tau} N^k$$

where τ is the step size and N^k is a vector of standard i.i.d Gaussian noise.

- Sufficient decrease condition [Armijo '66]. Set $\tau > 0$ such that

$$E(x^k) - E(x^{k+1}) - \sigma\tau \left\| \nabla E(x^k) \right\|^2 \geq 0,$$

where $\sigma \in (0, 1)$.

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- The Metropolis-Hastings rule [Metropolis et al. '53][Hastings '70]: Accept x^{k+1} if

$$\min \left\{ 0, E(x^k) - E(x^{k+1}) - \tau \left\| \nabla E(x^{k+\frac{1}{2}}) \right\|^2 - \frac{\sqrt{2\tau}}{2} \left\langle \nabla E(x^{k+\frac{1}{2}}), N^k \right\rangle \right\} > \log u,$$

where $u \sim \mathcal{U}[0, 1]$ and $\nabla E(x^{k+\frac{1}{2}}) = (\nabla E(x^k) + \nabla E(x^{k+1}))/2$ is the average gradient.

MAP vs. MMSE

- Application of GD and ULA to total variation regularized image denoising:

$$R(x) = \lambda \sum_{i,j} |x_{i+1,j} - x_{i,j}|_{\varepsilon} + |x_{i,j+1} - x_{i,j}|_{\varepsilon}, \quad D(x, y) = \frac{1}{2\sigma^2} \|x - y\|^2,$$

using $\varepsilon = 10^{-3}$, $\lambda_{MAP} = 10$, $\lambda_{MMSE} = 20$, $\sigma = 0.1$, step size $\tau = 2/L$.

MAP solution

MMSE solution

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- ▶ **Sampling:** Despite clear structural and algorithmic similarity, the progress in developing faster sampling algorithms seems to be much slower.
- ▶ **Question:** Can we leverage ideas from optimization to develop faster sampling algorithms?

Half-quadratic minimization

- ▶ A fruitful idea in optimization has always been lifting the objective function to a high-dimensional space where the new representation is expected to offer a better structure.
- ▶ Let us consider a popular technique called **half-quadratic minimization** [Geman, Reynolds, '92], which expresses a function $f(x)$ as

$$f(x) = \min_z q(x, z)$$

where q is quadratic in x .

- ▶ The lifting represents the function $f(x)$ as the infimum over a family of quadratic functions.
- ▶ Minimizing $f(x)$ is replaced by **alternating minimization** with respect to x and z .

$$x^{k+1} \in \operatorname{argmin}_x q(x, z^k), \quad z^{k+1} = \operatorname{argmin}_z q(x^{k+1}, z).$$

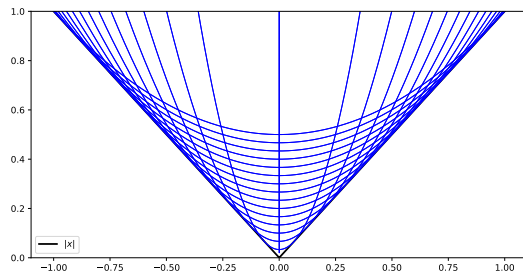
- ▶ Many variants exist (multiplicative, additive) and there are also relations to the convex conjugate [Nikolova, Ng '05].

Minimum envelope

- ▶ The total variation regularization term from the image denoising application can be written as a half-quadratic minimization problem [Chambolle, Lions '97]
- ▶ It is based on rewriting the absolute function as

$$R(x) = \lambda \sum_{i,j} |x_{i+1,j} - x_{i,j}| + |x_{i,j+1} - x_{i,j}|, \quad |t| = \min_{z>0} \frac{|t|^2}{2z} + \frac{z}{2},$$

which is quadratic in x and simple in z with $z^*(t) = |t|$.



Example

- ▶ For the TV denoising example the half-quadratic minimization algorithm converges within a few iterations.

Gibbs sampling

- Consider the following lifting of the prior p_X with latent variables z :

$$p_X(x) = \int p_{X,Z}(x, z) \, dz,$$

where $p_{X,Z}$ is expected to have a better structure for sampling.

- The sampling analogue to alternating minimization is **Gibbs sampling** [Geman, Geman, '84]

$$x^{k+1} \sim p_{X|Z=z^k}, \quad z^{k+1} \sim p_{Z|X=x^{k+1}},$$

which alternates sampling from the conditional distributions.

- Gibbs sampling can be shown to be a particular instance of the Metropolis-Hastings algorithm [Metropolis et al. '53][Hastings '70].

Gaussian latent machine (GLM)

- Assume the following **Product of Experts (PoE)** prior model [Hinton '99]

$$p_X(x) \propto \prod_{j=1}^m \phi_j((Kx)_j),$$

where $K : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear operator and $\phi_j : \mathbb{R} \rightarrow \mathbb{R}^+$ are 1D factors.

- In case K is a convolutional operator, the model is equivalent to the **Fields of Experts (FoE)** prior model [Roth, Black '05].
- In [Kuric, Zach, Habring, Unser, P. '25] we show that a PoE prior admits the following lifted representation in the form of a **Gaussian latent machine (GLM)**

$$p_{X,Z}(x, z) \propto \prod_{j=1}^m \mathcal{N}((Kx)_j | \mu_j(z_j), \sigma_j^2(z_j)) p_j(z_j) = \underbrace{\mathcal{N}(Kx | \tilde{\mu}(z), \tilde{\Sigma}(z))}_{p_{X|Z=z}(x)} \cdot \prod_{j=1}^m p_j(z_j),$$

where $\tilde{\mu}(z) = (\tilde{\mu}_1(z_1), \dots, \tilde{\mu}_m(z_m))$ and $\tilde{\Sigma}(z) = \text{diag}(\tilde{\sigma}_1^2(z_1), \dots, \tilde{\sigma}_m^2(z_m))$.

The conditional distributions

- ▶ For Gibbs sampling we need to have access to the conditional distributions.
- ▶ The conditional distribution $p_{X|Z}$ is simply given by the multivariate Gaussian

$$p_{X|Z}(x, z) = \mathcal{N}(Kx | \tilde{\mu}(z), \tilde{\Sigma}(z)) = \mathcal{N}(x | \mu(z), \Sigma(z))$$

where

$$\mu(z) = (K^\top \tilde{\Sigma}(z)^{-1} K)^{-1} K^\top \tilde{\Sigma}(z)^{-1} \tilde{\mu}(z), \quad \Sigma(z) = (K^\top \tilde{\Sigma}(z)^{-1} K)^{-1}.$$

- ▶ The conditional distribution $p_{X|Z}$ decomposes into m independent univariate distributions

$$p_{Z|X} = \prod_{j=1}^m p_{Z_j|X}, \quad p_{Z_j|X=x}(z_j) \propto \mathcal{N}((Kx)_j | \mu_j(z_j), \sigma_j^2(z_j)) \cdot p_j(z_j),$$

where p_j basically depends on the factors ϕ_j .

- ▶ In many situations of practical interest, e.g. 1D Gaussian mixture models, there are closed form solutions for the univariate distributions $p_{Z_i|X}$

- Consider again the 2D total variation image prior

$$p_X(x) \propto \prod_{i,j} \exp(-\lambda |x_{i+1,j} - x_{i,j}|) \exp(-\lambda |x_{i,j+1} - x_{i,j}|).$$

- Next, we consider a lifted representation of 1D Laplacian factors $\phi(t) = \exp(-\lambda|t|)$ as a Gaussian scale mixture

$$\frac{\lambda}{2} \exp(-\lambda|t|) = \int_{\mathbb{R}^+} \frac{1}{\sqrt{2\pi z}} \exp\left(-\frac{t^2}{2z}\right) \frac{\lambda^2}{2} \exp\left(-\lambda^2 \frac{z}{2}\right) dz$$

- From the Gaussian component, one directly sees that $\mu(z) = 0$ and $\sigma^2(z) = z$.
- The 1D conditional distribution $p_{Z_i|X=t}$ is given by a generalized inverse Gaussian

$$p_{Z_i|X=t} \propto z^{-\frac{1}{2}} \exp\left(-\left(\frac{t^2}{z} + \lambda^2 z\right)/2\right),$$

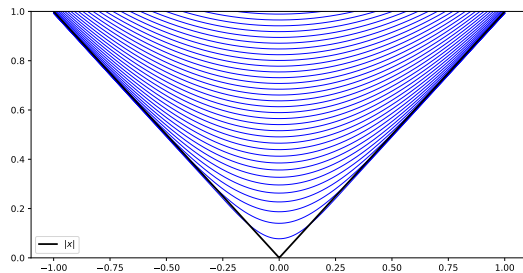
from which sampling is “relatively” easy.

Soft half-quadratic minimization

- In the negative log-domain it turns out that we obtain a soft-minimum over quadratic functions

$$\lambda|t| = -\log \left(\int_{\mathbb{R}^+} \frac{2}{\sqrt{2\pi}z} \exp \left(- \underbrace{\left(\frac{t^2}{2z} + \frac{\lambda^2 z}{2} \right)}_{q(t,z)} \right) dz \right),$$

where $q(t,z)$ is indeed quadratic in t .



Gibbs sampling on the Gaussian latent machine

- The Gibbs sampler to sample from p_X alternates sampling from a multivariate Gaussian and m independent generalized inverse Gaussians

$$\begin{cases} x^{k+1} \sim \mathcal{N}(x|0, (K^\top \text{diag}(z_1^k, \dots, z_m^k)^{-1} K)^{-1}), \\ z_j^{k+1} \sim GIG(\lambda^2, (Kx^{k+1})_j^2, \frac{1}{2}), j = 1 \dots m \end{cases}$$

- Sampling from the prior $p_X(x)$ can be easily extended to sampling from the posterior $p_{X|Y} = p_{Y|X} \cdot p_X$ by combining with a Gaussian likelihood term for denoising

$$p_{Y|X=x}(y) = \mathcal{N}(y|x, \sigma^2 \cdot I)$$

which only modifies the Gaussian in the first step of the Gibbs algorithm.

$$\begin{cases} x^{k+1} \sim \mathcal{N}(x|0, (K^\top \text{diag}(z_1^k, \dots, z_m^k)^{-1} K)^{-1}) \cdot \mathcal{N}(y|x, \sigma^2 \cdot I), \\ z_j^{k+1} \sim GIG(\lambda^2, (Kx^{k+1})_j^2, \frac{1}{2}), i = 1 \dots m \end{cases}$$

Sampling from the TV denoising posterior

- ▶ The MMSE computed from the samples of the GLM converges in a few iterations.

Sampling from a Gaussian vs. quadratic minimization

- ▶ Consider the following multivariate Gaussian distribution

$$\pi_X(x) \propto \exp\left(-\frac{1}{2}(x - \mu)^\top Q(x - \mu)\right),$$

- ▶ **Sampling:** Generate i.i.d. samples from the distribution
- ▶ **Optimization:** Find the mode of the distribution that is minimizing the quadratic function

$$\min_x \frac{1}{2}x^\top Qx - b^\top x.$$

- ▶ In optimization we assume that we only have access to $b = Q\mu$.

Gibbs vs. Gauss-Seidel

- ▶ Let us assume the following two-block structure

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \quad Q = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} Q_{11}\mu_1 + Q_{12}\mu_2 \\ Q_{21}\mu_1 + Q_{22}\mu_2 \end{bmatrix}$$

- ▶ A Gibbs sampling algorithm that takes advantage of the two-block structure is given by

$$\begin{cases} X_1^{k+1} \sim \mathcal{N}(\mu_1 - Q_{11}^{-1}Q_{12}(X_2^k - \mu_2), Q_{11}^{-1}), \\ X_2^{k+1} \sim \mathcal{N}(\mu_2 - Q_{22}^{-1}Q_{21}(X_1^{k+1} - \mu_1), Q_{22}^{-1}), \end{cases}$$

- ▶ A Gauss-Seidel (alternating minimization) algorithm takes the form

$$\begin{cases} x_1^{k+1} = Q_{11}^{-1}(b_1 - Q_{12}x_2^k) = \mu_1 - Q_{11}^{-1}Q_{12}(x_2^k - \mu_2), \\ x_2^{k+1} = Q_{22}^{-1}(b_2 - Q_{21}x_1^{k+1}) = \mu_2 - Q_{22}^{-1}Q_{21}(x_1^k - \mu_1), \end{cases}$$

- ▶ **Gibbs is just noisy Gauss-Seidel!**

Overrelaxation

- It is well known that the Gauss-Seidel algorithm can be accelerated using the method of successive overrelaxation (SOR) [Frankel '50]

$$\begin{cases} x_1^{k+1} = (1 - \omega)x_1^k + \omega Q_{11}^{-1}(b_1 - Q_{12}x_2^k), \\ x_2^{k+1} = (1 - \omega)x_2^k + \omega Q_{22}^{-1}(b_2 - Q_{21}x_1^{k+1}), \end{cases}$$

where $\omega \in (0, 2)$ is the relaxation parameter.

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where $\omega \in (0, 2)$ is the relaxation parameter.

- [Adler '81] observed that the Gibbs sampling algorithm can also be overrelaxed

$$\begin{cases} X_1^{k+1} \sim \mathcal{N}((1 - \omega)X_1^k + \omega(\mu_1 - Q_{11}^{-1}Q_{12}(X_2^k - \mu_2)), \omega(2 - \omega)Q_{11}^{-1}), \\ X_2^{k+1} \sim \mathcal{N}((1 - \omega)X_2^k + \omega(\mu_2 - Q_{22}^{-1}Q_{21}(X_1^{k+1} - \mu_1)), \omega(2 - \omega)Q_{22}^{-1}), \end{cases}$$

where it is crucial to rescale the covariance matrix by the factor $\omega(2 - \omega)$ in order to ensure the correct covariance of X^k .

- [Fox and Parker '17] proved a convergence rate, matching that of SOR.
- Values of $\omega \rightarrow 2$ result in negative autocorrelation of successive samples.

- Let's consider again the unadjusted Langevin algorithm (ULA)

$$X^{k+1} = X^k - \tau \nabla E(X^k) + \sqrt{2\tau} N^k \iff X^{k+1} \sim \mathcal{N}(X^k - \tau \nabla E(X^k), 2\tau \cdot I)$$

- Using the following splitting trick [Falk, Habring, P. '25]

$$X^{k+1} = \underbrace{X^k - \tau \nabla E(X^k) + \sqrt{\tau_1} N_1^k}_{Y^{k+1}} + \sqrt{\tau_2} N_1^k, \quad \tau_1 + \tau_2 = 2\tau,$$

we can interpret ULA as a two-block Gibbs algorithm

$$\begin{cases} Y^{k+1} \sim \mathcal{N}(X^k - \tau \nabla E(X^k), \tau_1 \cdot I), \\ X^{k+1} \sim \mathcal{N}(Y^{k+1}, \tau_2 \cdot I), \end{cases}$$

which however is not a true Gibbs algorithm because of a lack of a joint distribution in X and Y .

Inertial Langevin algorithm (ILA)

- Now let's apply the overrelaxation to the obtained Gibbs sampling scheme

$$\begin{cases} Y^{k+1} \sim \mathcal{N}((1-\omega)Y^k + \omega(X^k - \tau \nabla E(X^k)), \omega(2-\omega)\tau_1 \cdot I), \\ X^{k+1} \sim \mathcal{N}((1-\omega)X^k + \omega Y^{k+1}, \omega(2-\omega)\tau_2 \cdot I), \end{cases}$$

- The scheme can be reduced back to a single-variable scheme

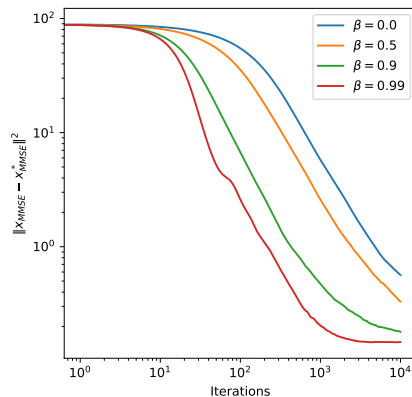
$$X^{k+1} = X^k - \gamma \nabla E(X^k) + \beta(X^k - X^{k-1}) + \sqrt{2\gamma(1-\beta)}N^k,$$

where we have defined $\gamma = \omega^2 \tau$ and $\beta = (1-\omega)^2$.

- The obtained scheme, which we term **Inertial Langevin Algorithm (ILA)** is nothing than the sampling analogue of the heavy ball algorithm [Polyak '64].
- It is crucial to scale the noise by the factor $\sqrt{2\gamma(1-\beta)}$.

Comparison

- ▶ The ground truth solution x_{MMSE}^* is computed using GLM.
- ▶ Choosing an inertial parameter $\beta \rightarrow 1$ leads to a significant acceleration.
- ▶ A small bias remains but can be accounted for using a Metropolis-Hastings acceptance test.



Underdamped Langevin

- ▶ The discrete ILA scheme can be shown to be a discretization of the underdamped Langevin SDE

$$\begin{cases} d\tilde{V}_t &= (-\delta\tilde{V}_t - \nabla E(\tilde{X}_t)) dt + \sqrt{2\delta} dW_t. \\ d\tilde{X}_t &= \tilde{V}_t dt, \end{cases}$$

with friction parameter $\delta > 0$, which is known to have a stationary distribution

$$\pi(x, v) \propto \exp\left(-\left(E(x) + \frac{\|v\|^2}{2}\right)\right)$$

- ▶ For a proof of convergence, we consider a slightly different parametrization of the ILA scheme, which allows to control the discretization error with required order.
- ▶ **Theorem:** Assume E is strongly convex with Lipschitz continuous gradient, then the sample distribution of the time-discrete ILA scheme converges to the stationary distribution of the underdamped Langevin SDE in $\mathcal{W}_2$ distance as $\Delta t \rightarrow 0$.

Conclusion

- ▶ There is an ongoing transition from pure optimization algorithms to more general sampling algorithms.
- ▶ Boils down to a difference between **min** and **softmin**, where the latter is more general.
- ▶ Many tricks that work well in optimization can be transformed to sampling.
 - ▶ Half-quadratic minimization $\rightsquigarrow$ Gaussian latent machine (GLM)
 - ▶ Inertial/accelerated gradient descent $\rightsquigarrow$ Inertial Langevin algorithm (ILA)
- ▶ Allows to perform Bayesian inference and uncertainty quantification for high-dimensional problems.
- ▶ For details and convergence rates see our recent preprints (will be online soon)

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Thanks for listening!

Appendix: Sampling from the multivariate Gaussian in the GLM

- In the GLM, we need to sample from a multivariate Gaussian with negative log density (up to constants)

$$\frac{1}{2}x^T \left(K^T \text{diag} \left(z_1^k, \dots, z_m^k \right)^{-1} K \right) x + \frac{1}{2\sigma^2} \|x - y\|^2,$$

where z are the latent variables.

- This quadratic function be rewritten in the standard form (up to constants)

$$\frac{1}{2}(x - \mu)^T (A^T A) (x - \mu)$$

with

$$A = \begin{bmatrix} \text{diag} \left(z_1^k, \dots, z_m^k \right)^{-\frac{1}{2}} K \\ \frac{1}{\sigma} \cdot I \end{bmatrix}, \quad \mu = \left(A^T A \right)^{-1} y / \sigma^2.$$

Sampling from a Gaussian with precision $A^\top A$

- ▶ The task is to sample from a Gaussian with negative log density

$$\frac{1}{2}(x - \mu)^\top (A^\top A) (x - \mu)$$

- ▶ A sample x is computed from a standard Gaussian sample $z \sim \mathcal{N}(0, I)$ as

$$x = \mu + (A^\top A)^{-1} A^\top z.$$

- ▶ A direct computation shows that

$$\text{cov}[x] = \left((A^\top A)^{-1} A^\top \right) \left((A^\top A)^{-1} A^\top \right)^\top = (A^\top A)^{-1}$$

Putting together

- Consider the previous equation

$$x = \mu + \left(A^\top A\right)^{-1} A^\top z$$

which when inserting the formula for them mean μ gives

$$x = \left(A^\top A\right)^{-1} y/\sigma^2 + \left(A^\top A\right)^{-1} A^\top z.$$

- Multiplying from the left with $A^\top A$ we obtain the linear system of equations

$$\left(A^\top A\right) x = y/\sigma^2 + A^\top z,$$

which can be solved using an iterative solver such as CG.